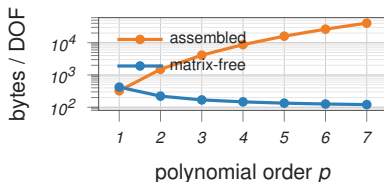


Matrix-free high-order CVFEM: past the assembly wall

The assembled wall

- Assembled CVFEM storage explodes with order p :
 $\text{nnz/row} \sim (2p+1)^3$
- Bytes/DOF grow $324 \rightarrow 40,500$ from $p=1$ to $p=7$
- Assembly infeasible beyond $p \approx 2-3$

Memory per DOF vs. polynomial order



Matrix-free answer

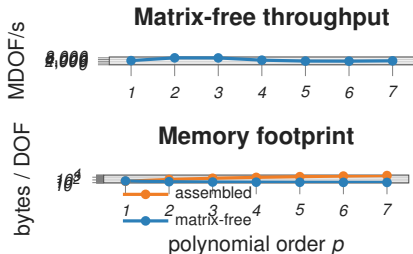
- High-order matrix-free CVFEM (Knaus / Nalu-Wind scheme), GPU-native in MARS
- Trilinos-free**: cstone octree + own CSR + cuSPARSE + Hypre
- Store per-element geometry *once*: recompute the $O(p^4)$

Crossover at $p=1$; for $p \geq 2$ matrix-free is the only affordable option.

Performance & memory vs. the assembled kernel

Single-GPU H100 order sweep

- Throughput **flat with order**:
4.3–8.0 GDOF/s ($p=7 \approx p=1$)
- Memory *drops* $421 \rightarrow 121$ B/DOF; assembled up to **336 $\times$** larger at $p=7$
- $p=1$: assembled SpMV wins one matvec (9.2 vs 6.0 GDOF/s); $p \geq 2$ matrix-free is the *only* option, no penalty
- Kernel: **75%** occupancy, ~ 2 TB/s ($\approx 60\%$ peak), L1-bound; FP64 tensor cores measured – *don't* help

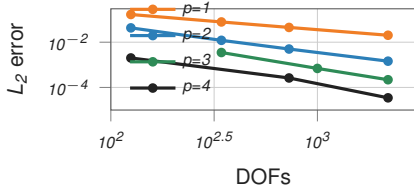


Accuracy & validation

High order pays off

- Convergence reproduces the reference (paper Table 1):
 $\sim p+1$ order
- At a fixed $\sim 2,200$ DOFs, $p=4$ is $\sim 570\times$ more accurate than $p=1$

L_2 error vs. DOFs (slope $\approx p+1$)



Validation

- GPU apply **bit-exact** vs an independent host reference, $p=1..7$ (machine precision)
- Consistent: annihilates linears on sheared / distorted meshes
- Verified faithful to the Knaus paper's Algorithm 2

Steepening slopes confirm $p+1$ convergence; the vertical gap at fixed DOFs is the high-order payoff.

Scaling & outlook

Toward trillions of DOFs

- Weak scaling on block hex meshes block_2M → block_64M on GH200 (**runs in progress**)
- Procedural per-rank mesh gen – no mesh file
- Path to 10^9 – 10^{12} DOF (~ 1550 GH200 at the 647M-DOF/GPU $p=4$ ceiling)

Differentiator

- Native **unstructured AMR** on the cstone octree – Nalu-Wind / Kynema-UGF use structured-AMR overset

Next: the solve (FlexGMRES + LOR-AMG) · conforming-unstructured 10^9 DOF via uniform refinement of an unstructured base

